

# MOSIP Decode 2026 — Problem Statement 1

Automated Conformance Testing for Inji Certify & Inji Verify

Findings Log and Open Questions

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## Abstract

This document records, day by day, the conformance testing work carried out against the Inji stack as part of MOSIP Decode Problem Statement 1. It captures what was run, what failed, the evidence for each failure, and the questions we need answered by MOSIP mentors. Every finding is tagged as **CONFIRMED** or **UNVERIFIED**; we do not report a defect upstream until it is reproducible and the root cause is understood.

## Contents

|          |                                                                             |          |
|----------|-----------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Scope and System Under Test</b>                                          | <b>2</b> |
| <b>2</b> | <b>Day Log</b>                                                              | <b>3</b> |
| 2.1      | Day 1 — Saturday, 20 September 2026 . . . . .                               | 3        |
| <b>3</b> | <b>Findings Catalogue</b>                                                   | <b>5</b> |
| F-01     | — Nonce has insufficient entropy . . . . .                                  | 5        |
| F-02     | — Nonce contains non-URL-safe characters . . . . .                          | 5        |
| F-03     | — No DCQL support; test plan cannot proceed . . . . .                       | 5        |
| F-04     | — Uses superseded <code>presentation_definition</code> . . . . .            | 6        |
| F-05     | — <code>client_metadata</code> absent in inline mode . . . . .              | 6        |
| F-06     | — No <code>direct_post.jwt</code> ; HAIP certification impossible . . . . . | 6        |
| F-07     | — DID document verification method identifier inconsistency . . . . .       | 7        |
| F-08     | — No mdoc / ISO 18013-5 support . . . . .                                   | 7        |
| F-09     | — Client identifier prefix has no overlap with 1.0 Final . . . . .          | 7        |
| <b>4</b> | <b>Open Questions for Mentors</b>                                           | <b>8</b> |
| <b>5</b> | <b>Reproduction</b>                                                         | <b>9</b> |

# 1 Scope and System Under Test

## Problem statement

Build an automated harness that spins up Inji Certify and Inji Verify, drives the OpenID Foundation conformance suite’s REST API programmatically to run the issuer and verifier test plans, and feeds the results into each module’s existing `api-testrig`, producing benchmark-gated reports with CI/CD integration.

## Components under test

| Component             |             | Version             | Notes                                                                  |
|-----------------------|-------------|---------------------|------------------------------------------------------------------------|
| Inji Verify (service) |             | 0.18.2              | <code>injistack/inji-verify-service</code>                             |
| Inji Verify (UI)      |             | 0.18.2              | <code>injistack/inji-verify-ui</code> , amd64 under emulation on arm64 |
| Inji Certify          |             | 0.14.0              | Not yet exercised (deferred, see Plan)                                 |
| OpenID Suite          | Conformance | 5.3.1 (rev 440eec8) | Local Docker deployment, prebuilt images                               |

## Test environment

- Host: Apple Silicon (arm64) macOS, Docker Desktop.
- Conformance suite reachable at `https://localhost.emobix.co.uk:8443` (self-signed certificate; hostname resolves publicly to 127.0.0.1).
- Inji Verify exposed publicly via an ngrok HTTPS tunnel. A public hostname is mandatory because Verify identifies itself with a `did:web` DID, which is resolved over public HTTPS.
- The conformance suite’s REST API required no authentication token in the local `dev` profile, consistent with the problem statement.

## Evidence handling

Conformance suite logs are exported as signed JSON (`.json` plus `.json.sig`) and stored under `logs/<date>/`. Findings below cite the exact log line (`src` field) and the OID4VP specification section reported by the suite.

## 2 Day Log

### 2.1 Day 1 — Saturday, 20 September 2026

#### Objective

Stand up the conformance suite and one MOSIP component, and complete one manual conformance test end to end to establish a baseline.

#### Decisions taken

1. **Inji Verify before Inji Certify.** Verify's Docker Compose brings up three containers with no external dependencies. Certify's `docker-compose-injistack` requires a PKCS12 keystore obtained through an onboarding process, a public DID endpoint, an external authorisation server, and partner API keys. The verifier test plan also has 12 modules against the issuer plan's 61. Verify is the cheaper path to a first result, and everything learned transfers.
2. **Non-HAIP plan first.** The HAIP verifier plan offers only `direct_post.jwt` as a response mode. Inji Verify hardcodes `direct_post` (see F-06), so no HAIP test can complete. We ran *OpenID for Verifiable Presentations 1.0 Final: Test a verifier — alpha tests* instead, which offers plain `direct_post`.
3. **Credential format `sd_jwt_vc`.** The plan offers only `sd_jwt_vc` and `iso_mdl`. Verify has no mdoc support whatsoever (F-08), making `iso_mdl` unusable.

#### Test plan configuration

- Plan: `oid4vp-1final-verifier-test-plan`, plan ID `1N7C4DH1HvYmc`
- Variant: `credential_format=sd_jwt_vc, client_id_prefix=redirect_uri, request_method=url_query, vp_profile=plain_vp, response_mode=direct_post`
- Alias: `injiverify-shardul`

#### Runs executed

| Test ID          | Outcome              | Notes                                                                                                                                                                                             |
|------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dgeeosZpIfDK15   | FAILED / INTERRUPTED | First attempt. Request object had expired (5-minute window). Discarded as an artefact of manual timing, not a defect.                                                                             |
| dMg4B5whxWgShFA  | FAILED / INTERRUPTED | Fresh request (2m05s old). Suite never dereferenced <code>request_uri</code> because the <code>url_query</code> variant expects inline parameters. Root cause understood; see F-09.               |
| MtzipWoD2dVgtiFA | FAILED / INTERRUPTED | After switching the credential's <code>clientIdScheme</code> to <code>pre_registered</code> . Reached 20+ checks; produced the substantive findings below. <b>This is the Day 1 baseline run.</b> |

#### Configuration change made

In `inji-verify/docker-compose/config/config.json`, the credential *Mock Identity (SD JWT)* was changed from `"clientIdScheme":"did"` to `"clientIdScheme":"pre_registered"`. Per Inji Verify's own README, `did` delivers the authorisation request *by reference* via `request_uri`, whereas `pre_registered` delivers it *inline*. Only the inline form matches the `request_method=url_query` variant. Backup retained at `config.json.bak`.

**Result of the baseline run (MtzpWoD2dVgtic0)**

Approximately fourteen checks passed, including nonce extraction, response type, response mode, absence of `client_id_scheme`, absence of `scope`, absence of `redirect_uri`, response URI validity, and the outer parameter allow-list. Three checks failed and three raised warnings. The test terminated at `ExtractDCQLQueryFromAuthorizationRequest`.

**Status at end of Day 1**

Test 01 (`happy-flow`) complete and documented. Tests 02–11 deferred to Day 2.

### 3 Findings Catalogue

Severity reflects impact on certifiability and on security, not on our convenience.

#### F-01 — Nonce has insufficient entropy HIGH CONFIRMED

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final, §5.2

**Observation.** The nonce in Verify’s OID4VP authorisation request is a base64-encoded millisecond Unix timestamp rather than a cryptographically random value.

**Evidence — our analysis.** Three samples decoded, across both request-construction paths:

| clientIdScheme | nonce (base64)       | decoded       | $\Delta$ to issue-dAt |
|----------------|----------------------|---------------|-----------------------|
| did            | MTc4OTg4MzIxMzQ5MA== | 1789883213490 | —                     |
| pre_registered | MTc4OTg4NDUzNjY3OQ== | 1789884536679 | 120 ms                |
| pre_registered | MTc4OTg4NDY1NjQzNg== | 1789884656436 | 158 ms                |

**Evidence — conformance suite, independently.**

VP1FinalEnsureMinimumNonceEntropy — WARNING

“Nonce does not appear to have sufficient entropy (calculated Shannon entropy is below the threshold). Section 5.2 of OID4VP 1.0 Final requires the nonce to be a ‘fresh, cryptographically random number with sufficient entropy’; OpenID4VP PR #722 recommends at least 128 bits.”

actual: 73.68 expected: 96.0

**Impact.** The nonce exists to bind a wallet’s presentation to a specific request and prevent replay. A value derived from the current clock is predictable to within milliseconds, substantially weakening that guarantee. In a national identity deployment this is a meaningful security property, not a cosmetic issue.

**Proposed action.** File upstream against mosip/inji-verify, recommending a CSPRNG-generated nonce of at least 128 bits, base64url-encoded without padding (which also resolves F-02).

#### F-02 — Nonce contains non-URL-safe characters MEDIUM CONFIRMED

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final, §5.2

**Evidence.**

CheckForInvalidCharsInNonce — FAILURE

“Non URL safe characters found in nonce. This may introduce interoperability issues.”

invalid\_chars: ["="]

**Observation.** The nonce uses standard base64 with = padding rather than base64url. The = character is not URL-safe and requires percent-encoding when the nonce travels in a query string, which not all wallets handle consistently.

**Impact.** Interoperability risk with third-party wallets. Same root cause as F-01 and fixed by the same change.

#### F-03 — No DCQL support; test plan cannot proceed HIGH CONFIRMED

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final, §6

**Evidence — conformance suite.**

ExtractDCQLQueryFromAuthorizationRequest — FAILURE

“dcql\_query not found in authorization request parameters”

**Evidence — source.** A case-insensitive search for `dcql` across `verify-service/src/main` returns zero matches. Credential selection is implemented entirely via `presentation_definition` with `input_descriptors` (DIF Presentation Exchange), set in `VerifiablePresentationRequestServiceImpl` as the `presentation_definition_uri` claim.

**Impact.** This terminates the test. Because every module in the 1.0 Final verifier plan requires a DCQL query, **Inji Verify 0.18.2 cannot pass any test in that plan.** This is the single most consequential finding of Day 1 and is not fixable by configuration.

**Open question for mentors.** See Q1.

#### F-04 — Uses superseded `presentation_definition` **MEDIUM** **CONFIRMED**

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final

**Evidence.**

`CheckNoPresentationDefinitionInVpAuthorizationRequest` — WARNING  
 “Authorization request contains legacy ‘`presentation_definition`’. VP1 Final uses ‘`dcql_query`’ for credential selection, so sending both is unnecessary and may indicate draft and final request formats are being mixed.”

**Relationship to F-03.** Two faces of the same gap: Verify implements the earlier draft’s query model. Recorded separately because a maintainer may address them independently (e.g. support both during a transition period).

#### F-05 — `client_metadata` absent in inline mode **MEDIUM** **UNVERIFIED**

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final, §5.1; Annex B.2.2, B.3.4

**Evidence.**

`VP1FinalValidateVpFormatsSupportedInClientMetadata` — FAILURE  
 “`client_metadata` is missing or not a JSON object”  
`VP1FinalCheckForUnexpectedParametersInVpClientMetadata` — WARNING  
 “`client_metadata` not present in request but it is mandatory to include `vp_formats` in it”

**Why this is not yet confirmed.** In the `did` (by-reference) path, Verify’s signed request object `did` contain `client_metadata` with a full `vp_formats` block advertising `ldp_vp` and `vc+sd-jwt`. In the `pre_registered` (inline) path, the `/vp-session-request` API response contained no `client_metadata` field, so the URI we constructed could not include one. It is not yet established whether Verify omits it in inline mode, or whether the UI adds it when building the QR code and our reconstruction simply dropped it.

**Action required before reporting.** Inspect the QR payload the Verify UI actually produces in `pre_registered` mode and compare against the API response.

#### F-06 — No `direct_post.jwt`; HAIP certification impossible **HIGH** **CONFIRMED**

**Component:** Inji Verify 0.18.2 **Spec:** OID4VP 1.0 Final / HAIP

**Evidence — source.**

```
verify-service/src/main/java/io/inji/verify/shared/Constants.java
public static final String RESPONSE_MODE = "direct_post";
```

This is a `static final` constant, not a configuration property or environment variable. No code path produces the encrypted `direct_post.jwt` variant.

**Evidence — conformance suite.** The HAIP verifier plan (*OpenID for Verifiable Presentations 1.0 Final/HAIP: Test a verifier*) offers only `direct_post.jwt` in its Response Mode variant.

**Impact.** Inji Verify cannot be submitted for HAIP self-certification in its current state. HAIP is the profile relevant to high-assurance national identity deployments, which is MOSIP’s own domain.

**Open question for mentors.** See Q2.

## F-07 — DID document verification method identifier inconsistency MEDIUM CONFIRMED

**Component:** Inji Verify 0.18.2    **Spec:** W3C DID Core

**Evidence.** Retrieved from `https://<host>/v1/verify/did.json`:

```
"id": "did:web:<host>:v1:verify"    (document subject)
"verificationMethod[0].id": "did:web:<host>:verify#key-0"  (no :v1:)
"assertionMethod[0]": "did:web:<host>:verify#key-0"      (no :v1:)
```

**Origin.** The default Docker Compose environment variables differ: `INJI_DID_VERIFY_URI` includes the `v1` path segment while `INJI_DID_VERIFY_PUBLIC_KEY_URI` does not.

**Impact.** Under `did:web` resolution rules, the key’s identifier resolves to `https://<host>/verify/did.json`, a path that does not exist on the deployment. A strict relying party resolving the verification method by its own identifier would fail.

## F-08 — No mdoc / ISO 18013-5 support LOW CONFIRMED

**Component:** Inji Verify 0.18.2

**Evidence.** Searches for `mso_mdoc`, `iso_md1` and `mdoc` across `verify-service/src/main` return zero matches.

**Impact.** The `iso_md1` credential format variant of the verifier plan cannot be exercised. Recorded for completeness; likely a deliberate scope decision rather than a defect.

## F-09 — Client identifier prefix has no overlap with 1.0 Final MEDIUM CONFIRMED

**Component:** Inji Verify 0.18.2    **Spec:** OID4VP 1.0 Final, §5.9.2

**Evidence.**

`EnsureClientIdMatchesResponseUri` — FAILURE

“For `redirect_uri` `client_id` prefix, the `client_id` (without prefix) must equal `response_uri`”

```
client_id: "inji-verify-ui"  response_uri: "https://<host>/v1/verify/vp-submission/direct-post"
```

**Observation.** Inji Verify supports two client identifier schemes, `did` and `pre_registered`. The 1.0 Final verifier plan offers three prefixes: `redirect_uri`, `x509_san_dns`, `x509_hash`. Verify has no X.509 material configured, so the two X.509 options are unreachable, and `redirect_uri` requires a client identifier Verify does not produce. **No variant combination in the plan matches Verify’s client identification model.**

**Caveat.** This failure is partly a consequence of the variant we selected. The reportable observation is the absence of overlap, not that any single value is “wrong”.

**Open question for mentors.** See Q3.

## 4 Open Questions for Mentors

These are the questions we would like to put to MOSIP mentors on the community forum or in a Weekly Connect session. Each is phrased to be answerable without our setup in front of them.

### Q1 — Target specification version.

Inji Verify 0.18.2 implements OID4VP credential selection via `presentation_definition` (Presentation Exchange), whereas OID4VP 1.0 Final requires `dcql_query`. Consequently no test in the 1.0 Final verifier plan can complete. Which specification version should our harness target as the benchmark: 1.0 Final (accepting that the current baseline is a full set of failures), or the ID2 draft that the shipped implementation appears to target? Is DCQL support on the Inji roadmap, and if so for which release?

### Q2 — HAIP certification intent.

The problem statement asks us to document “how to submit for self-certification”. The HAIP verifier plan mandates `direct_post.jwt`, but `RESPONSE_MODE` is a `static final` constant set to `direct_post` in `Constants.java`. Is encrypted response mode planned? Should our harness include the HAIP plan as a known-failing suite to track progress toward certification, or exclude it until the capability exists?

### Q3 — Client identifier scheme.

Verify supports `did` and `pre_registered` client identifier schemes; the 1.0 Final plan offers `redirect_uri`, `x509_san_dns` and `x509_hash`. Is there a supported configuration that allows an Inji Verify deployment to present a client identifier the 1.0 Final conformance plan can validate, or is this a known gap?

### Q4 — Benchmark semantics for `api-testrig`.

The problem statement asks for results “gated against a conformance benchmark”. Given that the honest current baseline is “all modules fail at DCQL extraction”, should the benchmark be (a) a frozen snapshot of today’s pass/fail set, so the gate catches regressions only, or (b) an aspirational target that the build fails against until conformance improves? Our current design assumes (a), with an expected-failures file under version control.

### Q5 — Sample configuration shipped with Inji Verify.

The default `config.json` ships five credential types, four of which use `ldp_vc` with `clientIdScheme: did`. That combination cannot be exercised by the 1.0 Final plan at all. Is the shipped sample configuration intended to be certification-representative, or purely a demonstration fixture? This affects whether we report the configuration itself as a finding.

### Q6 — Interactive tests and automation.

Every module in the verifier plan is interactive: it waits for a human to trigger a flow in the verifier and paste the resulting authorisation request, and the happy-flow module also requires a screenshot upload before it reaches `REVIEW` status. Our harness can drive request generation via Verify’s own API, but the screenshot-upload requirement appears to need `POST /api/log/{id}/images`. Is a fully unattended run of the verifier plan considered achievable, or is a documented semi-automated flow the expected deliverable?

### Q7 — Scope of Inji Certify work.

The `docker-compose-injistack` setup requires a PKCS12 keystore obtained through Mimoto onboarding, a publicly resolvable DID endpoint, and an external authorisation server. For a two-person team on a four-week timeline, is there a lighter-weight Certify configuration suitable for conformance testing, or is the Collab sandbox the intended target for the issuer plan?

## 5 Reproduction

To reproduce the Day 1 baseline run:

1. Follow `docs/SETUP.md` to install prerequisites and clone the repositories.
2. Follow `docs/RUNBOOK.md` to start the conformance suite, Inji Verify, and ngrok.
3. In `inji-verify/docker-compose/config/config.json`, ensure the *Mock Identity (SD JWT)* entry has `"clientIdScheme":"pre_registered"`.
4. In the conformance suite, create the plan *OpenID for Verifiable Presentations 1.0 Final: Test a verifier — alpha tests* with the variant recorded in Section 2.
5. Run module 01, `oid4vp-1final-verifier-happy-flow`.
6. In the Verify UI, request the *Mock Identity (SD JWT)* credential; capture the `/v1/verify/vp-session-requ` response.
7. Run `python3 scripts/build-oid4vp-uri.py` on that response and paste the resulting URI into the suite. The request expires 300 seconds after issue.
8. Export the signed log and store it under `logs/<date>/.`

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*Day 2 section to be appended after tests 02–11 are executed.*